P8131 Homework 7

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Problem 1

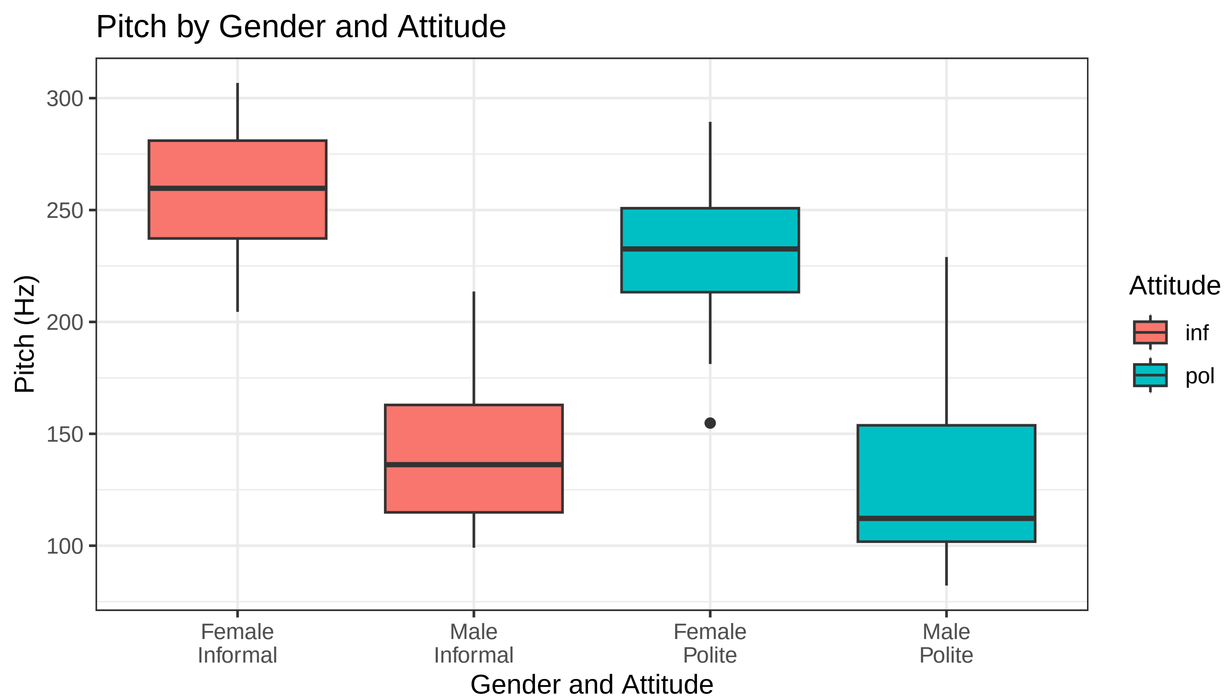
In a pitch study, we are interested in the relationship between pitch and politeness. There are two levels of politeness (a formal register: pol, and an informal register: inf). On top of that, we also have an additional fixed effect, gender. Each subject was tested on several scenarios (e.g., asking a peer for a favor (informal condition) or asking a professor for a favor (formal condition)). The pitch measurements are typically correlated for the same subject and in the same scenario.

```
pitch <- read.csv("HW7_politeness_data.csv")
pitch$subject <- factor(pitch$subject)
pitch$gender <- factor(pitch$gender)
pitch$attitude <- factor(pitch$attitude)
pitch$scenario <- factor(pitch$scenario)
```

(a) Exploratory analysis

Provide boxplots to show the relation between gender/attitude and pitch (ignoring different scenarios).

```
ggplot(pitch, aes(x = interaction(gender, attitude), y = frequency,
                  fill = attitude)) +
  geom_boxplot() +
  labs(x = "Gender and Attitude", y = "Pitch (Hz)",
       title = "Pitch by Gender and Attitude",
       fill = "Attitude") +
  scale_x_discrete(labels = c("Female\nInformal", "Male\nInformal",
                              "Female\nPolite", "Male\nPolite")) +
  theme_bw()
```



Females tend to have higher pitch than males across both attitude conditions. Within each gender, the polite condition tends to have slightly lower pitch than the informal condition, suggesting that speakers lower their pitch when being polite.

(b) Mixed effects model with random intercepts for subjects

Fit a mixed effects model with random intercepts for different subjects (gender and attitude being the fixed effects). What is the covariance matrix for a subject Y_i? What is the covariance matrix for the estimates of fixed effects (Hint: 3×3 matrix for intercept, gender and attitude)? What are the BLUPs for subject-specific intercepts? What are the residuals?

```r
LMM1 <- lme(frequency ~ gender + attitude, random = ~ 1 | subject,
            data = pitch, method = "REML")
summary(LMM1)
```

```
## Linear mixed-effects model fit by REML
##   Data: pitch
##        AIC      BIC    logLik
##   806.0805 818.0527 -398.0402
##
## Random effects:
##  Formula: ~1 | subject
##         (Intercept) Residual
## StdDev:    24.45803 29.11537
##
## Fixed effects:  frequency ~ gender + attitude
##                   Value Std.Error DF   t-value p-value
## (Intercept)  256.98690 15.154986 77 16.957251  0.0000
## genderM     -108.79762 20.956235  4 -5.191659  0.0066
## attitudepol  -20.00238  6.353495 77 -3.148248  0.0023
```

```
##  Correlation:
##             (Intr) gendrM
## genderM     -0.691
## attitudepol -0.210  0.000
##
## Standardized Within-Group Residuals:
##        Min         Q1        Med         Q3        Max
## -2.3564422 -0.5658319 -0.2011979  0.4617895  3.2997610
##
## Number of Observations: 84
## Number of Groups: 6
```

Covariance matrix for Y_i

This is a random intercept model: $Y_{ij} = \beta_0 + \beta_1 \cdot \text{gender}_i + \beta_2 \cdot \text{attitude}_{ij} + b_i + \epsilon_{ij}$, where $b_i \sim N(0, \sigma_b^2)$ and $\epsilon_{ij} \sim N(0, \sigma^2)$, independently.

From the random intercept model theory (Lecture 17, slide 6), the marginal covariance of Y_i has compound symmetry structure:

$$\text{Cov}(Y_i) = \begin{pmatrix} \sigma_b^2 + \sigma^2 & \sigma_b^2 & \cdots & \sigma_b^2 \\ \sigma_b^2 & \sigma_b^2 + \sigma^2 & \cdots & \sigma_b^2 \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_b^2 & \sigma_b^2 & \cdots & \sigma_b^2 + \sigma^2 \end{pmatrix}_{14 \times 14}$$

since each subject has $n_i = 14$ observations (7 scenarios $\times$ 2 attitudes).

```r
sigma_b <- as.numeric(VarCorr(LMM1)[1, 2]) # SD of random intercept
sigma_e <- as.numeric(VarCorr(LMM1)[2, 2]) # residual SD
sigma_b_sq <- sigma_b^2
sigma_e_sq <- sigma_e^2
cat("sigma_b^2 (random intercept variance):", round(sigma_b_sq, 2), "\n")
```

```
## sigma_b^2 (random intercept variance): 598.2
```

```r
cat("sigma^2 (residual variance):", round(sigma_e_sq, 2), "\n")
```

```
## sigma^2 (residual variance): 847.7
```

```r
cat("Diagonal entries:", round(sigma_b_sq + sigma_e_sq, 2), "\n")
```

```
## Diagonal entries: 1445.9
```

```r
cat("Off-diagonal entries:", round(sigma_b_sq, 2), "\n")
```

```
## Off-diagonal entries: 598.2
```

Plugging in the estimates, the 14×14 covariance matrix has diagonal entries $\hat{\sigma}_b^2 + \hat{\sigma}^2$ and off-diagonal entries $\hat{\sigma}_b^2$, with values shown above.

Covariance matrix for the fixed effects estimates

The 3×3 covariance matrix of $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2)^T$ is obtained via `vcov()`:

```
vcov(LMM1)
```

```
##             (Intercept)       genderM   attitudepol
## (Intercept)   229.67362 -2.195819e+02 -2.018345e+01
## genderM      -219.58189  4.391638e+02 -5.487266e-15
## attitudepol   -20.18345 -5.487266e-15  4.036690e+01
```

BLUPs for subject-specific intercepts

The BLUPs $\hat{b}_i$ represent each subject's predicted deviation from the population mean intercept:

```
random.effects(LMM1)
```

```
##    (Intercept)
## F1  -13.575831
## F2   10.170522
## F3    3.405309
## M3   27.960288
## M4    4.739325
## M7  -32.699613
```

Residuals

The within-subject residuals $\hat{\epsilon}_{ij} = Y_{ij} - \hat{Y}_{ij}$, where $\hat{Y}_{ij}$ includes both fixed effects and random intercepts:

```
LMM1$residuals
```

```
##          fixed     subject
## 1  -23.6845238 -10.1086926
## 2  -52.4869048 -38.9110735
## 3   48.1154762  61.6913074
## 4    2.7130952  16.2889265
## 5  -33.0845238 -19.5086926
## 6   29.9130952  43.4889265
## 7   13.8154762  27.3913074
## 8   19.8130952  33.3889265
## 9   -5.0845238   8.4913074
## 10  -4.5869048   8.9889265
## 11 -55.7845238 -42.2086926
## 12 -26.2869048 -12.7110735
## 13 -40.4869048 -26.9110735
## 14 -82.1845238 -68.6086926
## 15  -7.2845238 -10.6898326
## 16 -19.6869048 -23.0922136
## 17  -0.1845238  -3.5898326
## 18  -5.9869048  -9.3922136
## 19  30.0154762  26.6101674
```

```
## 20   9.0130952   5.6077864
## 21  38.4154762  35.0101674
## 22  49.8130952  46.4077864
## 23  -4.3845238  -7.7898326
## 24  -4.4869048  -7.8922136
## 25 -10.4845238 -13.8898326
## 26  21.8130952  18.4077864
## 27   7.4130952   4.0077864
## 28 -51.4845238 -54.8898326
## 29 -17.4869048 -22.2262298
## 30 -24.5892857 -29.3286108
## 31 100.8130952  96.0737702
## 32 -33.2892857 -38.0286108
## 33 -15.9869048 -20.7262298
## 34  65.4107143  60.6713892
## 35  65.2130952  60.4737702
## 36  14.7107143   9.9713892
## 37 -26.3869048 -31.1262298
## 38 -21.2892857 -26.0286108
## 39 -18.1869048 -22.9262298
## 40 -11.9892857 -16.7286108
## 41  -2.1892857  -6.9286108
## 42  -1.6869048  -6.4262298
## 43 -42.0869048  -9.3872916
## 44 -49.0892857 -16.3896725
## 45 -45.9869048 -13.2872916
## 46 -43.8892857 -11.1896725
## 47 -42.2869048  -9.5872916
## 48 -37.9892857  -5.2896725
## 49 -31.0869048   1.6127084
## 50 -28.1892857   4.5103275
## 51 -34.4869048  -1.7872916
## 52 -45.2892857 -12.5896725
## 53 -19.3869048  13.3127084
## 54 -39.9892857  -7.2896725
## 55 -23.7892857   8.9103275
## 56 -20.5869048  12.1127084
## 57  -4.2845238 -14.4550462
## 58 -25.6869048 -35.8574271
## 59   9.3154762  -0.8550462
## 60   2.7130952  -7.4574271
## 61  52.4154762  42.2449538
## 62  44.8130952  34.6425729
## 63   6.2154762  -3.9550462
## 64  39.2130952  29.0425729
## 65  40.7154762  30.5449538
## 66  37.2130952  27.0425729
## 67 -28.9845238 -39.1550462
## 68 -31.0869048 -41.2574271
## 69  24.0130952  13.8425729
## 70  -9.7845238 -19.9550462
## 71  25.6130952  -2.3471929
## 72  40.6107143  12.6504261
## 73  14.2130952 -13.7471929
```

```
## 74  51.5107143  23.5504261
## 75  32.0130952   4.0528071
## 76  37.9107143   9.9504261
## 77  79.3130952  51.3528071
## 78  42.7107143  14.7504261
## 79  32.5130952   4.5528071
## 80   8.3107143 -19.6495739
## 81  18.5130952  -9.4471929
## 82   9.8107143 -18.1495739
## 83  12.9107143 -15.0495739
## 84  25.1130952  -2.8471929
## attr(,"std")
##  [1] 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537
##  [9] 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537
## [17] 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537
## [25] 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537
## [33] 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537
## [41] 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537
## [49] 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537
## [57] 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537
## [65] 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537
## [73] 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537 29.11537
## [81] 29.11537 29.11537 29.11537 29.11537
```

(c) Model with interaction and likelihood ratio test

Fit a mixed effects model with intercepts for different subjects (gender, attitude and their interaction being the fixed effects). Use likelihood ratio test to compare this model with the model in part (b) to determine whether the interaction term is significantly associated with pitch.

For a valid likelihood ratio test comparing models with different fixed effects, both models must be fit using maximum likelihood (ML), not REML (Lecture 17, slide 20).

```r
# Refit both models with ML
LMM1.ml <- lme(frequency ~ gender + attitude, random = ~ 1 | subject,
               data = pitch, method = "ML")
LMM2.ml <- lme(frequency ~ gender * attitude, random = ~ 1 | subject,
               data = pitch, method = "ML")
summary(LMM2.ml)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: pitch
##        AIC      BIC    logLik
##   826.2508 840.8357 -407.1254
##
## Random effects:
##  Formula: ~1 | subject
##         (Intercept) Residual
## StdDev:    19.50493 28.67234
##
## Fixed effects:  frequency ~ gender * attitude
##                            Value Std.Error DF   t-value p-value
```

```
## (Intercept)          260.68571 13.200754 76 19.747790  0.0000
## genderM             -116.19524 18.668685  4 -6.224072  0.0034
## attitudepol          -27.40000  9.066991 76 -3.021951  0.0034
## genderM:attitudepol   14.79524 12.822662 76  1.153835  0.2522
##  Correlation:
##                     (Intr) gendrM atttdp
## genderM             -0.707
## attitudepol         -0.343  0.243
## genderM:attitudepol  0.243 -0.343 -0.707
##
## Standardized Within-Group Residuals:
##        Min         Q1        Med         Q3        Max
## -2.2856421 -0.5245601 -0.1718554  0.4929026  3.2293520
##
## Number of Observations: 84
## Number of Groups: 6
```

```r
lrt_result <- anova(LMM1.ml, LMM2.ml)
lrt_result
```

```
##         Model df      AIC      BIC    logLik   Test  L.Ratio p-value
## LMM1.ml     1  5 825.6363 837.7904 -407.8182
## LMM2.ml     2  6 826.2508 840.8357 -407.1254 1 vs 2 1.385523  0.2392
```

Hypotheses:

- H_0: $\beta_3 = 0$ (no gender-by-attitude interaction)
- H_1: $\beta_3 \neq 0$

The likelihood ratio test statistic is 1.3855 with 1 degree of freedom and a p-value of 0.2392. Based on the p-value from the test, the interaction between gender and attitude is not statistically significant at $\alpha = 0.05$, indicating that the effect of attitude on pitch does not differ significantly between genders.

(d) Crossed random effects model

Write out the mixed effects model with random intercepts for both subjects and scenarios (gender and attitude being the fixed effects). Fit the model using `lmer` in the `lme4` package. Write out the covariance matrix for a subject Y_i. What is the interpretation of the coefficient for the fixed effect term attitude?

Model specification

$$Y_{ijk} = \beta_0 + \beta_1 \cdot \text{gender}_i + \beta_2 \cdot \text{attitude}_k + b_i + c_j + \epsilon_{ijk},$$

where:

- $i = 1, \dots, 6$ indexes subjects, $j = 1, \dots, 7$ indexes scenarios, k indexes attitude level,
- $b_i \sim N(0, \sigma_b^2)$ is the random intercept for subject i,
- $c_j \sim N(0, \sigma_c^2)$ is the random intercept for scenario j,
- $\epsilon_{ijk} \sim N(0, \sigma^2)$ is the residual error,
- b_i, c_j, and ϵ_{ijk} are mutually independent.

Model fitting

```r
LMM3 <- lmer(frequency ~ gender + attitude + (1 | subject) + (1 | scenario),
             data = pitch)
summary(LMM3)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: frequency ~ gender + attitude + (1 | subject) + (1 | scenario)
##    Data: pitch
##
## REML criterion at convergence: 784.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.2690 -0.6331 -0.0878  0.5204  3.5326
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  scenario (Intercept) 224.5    14.98
##  subject  (Intercept) 613.2    24.76
##  Residual             637.8    25.25
## Number of obs: 84, groups:  scenario, 7; subject, 6
##
## Fixed effects:
##             Estimate Std. Error t value
## (Intercept)  256.987     16.101  15.961
## genderM     -108.798     20.956  -5.192
## attitudepol  -20.002      5.511  -3.630
##
## Correlation of Fixed Effects:
##             (Intr) gendrM
## genderM     -0.651
## attitudepol -0.171  0.000
```

Covariance matrix for Y_i

For subject i, Y_i is a 14×1 vector (7 scenarios $\times$ 2 attitudes). The covariance structure now depends on whether two observations share the same scenario:

- **Diagonal entries:** $\text{Var}(Y_{ijk}) = \sigma_b^2 + \sigma_c^2 + \sigma^2$.
- **Off-diagonal, same scenario** (same j, different attitude): $\text{Cov}(Y_{ijk}, Y_{ijk'}) = \sigma_b^2 + \sigma_c^2$ (both random effects shared).
- **Off-diagonal, different scenario** (different j, same subject): $\text{Cov}(Y_{ijk}, Y_{ij'k'}) = \sigma_b^2$ (only subject random effect shared).

```r
vc <- as.data.frame(VarCorr(LMM3))
sigma_b2 <- vc$vcov[vc$grp == "subject"]
sigma_c2 <- vc$vcov[vc$grp == "scenario"]
sigma_e2 <- vc$vcov[vc$grp == "Residual"]
cat("sigma_b^2 (subject):", round(sigma_b2, 2), "\n")
```

```
## sigma_b^2 (subject): 613.19
```

```r
cat("sigma_c^2 (scenario):", round(sigma_c2, 2), "\n")
```

```
## sigma_c^2 (scenario): 224.5
```

```r
cat("sigma^2 (residual):", round(sigma_e2, 2), "\n")
```

```
## sigma^2 (residual): 637.78
```

```r
cat("\nDiagonal entries:", round(sigma_b2 + sigma_c2 + sigma_e2, 2), "\n")
```

```
##
## Diagonal entries: 1475.47
```

```r
cat("Off-diagonal (same scenario):", round(sigma_b2 + sigma_c2, 2), "\n")
```

```
## Off-diagonal (same scenario): 837.69
```

```r
cat("Off-diagonal (different scenario):", round(sigma_b2, 2), "\n")
```

```
## Off-diagonal (different scenario): 613.19
```

If we order the observations within subject i by scenario (scenario 1: pol, inf; scenario 2: pol, inf; ...; scenario 7: pol, inf), the 14×14 covariance matrix has a block structure. Each 2×2 diagonal block (within a scenario) has diagonal $\sigma_b^2 + \sigma_c^2 + \sigma^2$ and off-diagonal $\sigma_b^2 + \sigma_c^2$. All entries in off-diagonal blocks (between different scenarios) equal σ_b^2.

Interpretation of the attitude coefficient

```r
fixef(LMM3)["attitudepol"]
```

```
## attitudepol
##   -20.00238
```

The coefficient for attitude (polite vs. informal as reference) has a population-average (marginal) interpretation: holding gender constant, the mean pitch is estimated to change by this amount (in Hz) when switching from the informal register to the polite register, averaged across all subjects and scenarios. A negative value indicates that, on average, speakers use a lower pitch in polite settings compared to informal settings.